

BTP Report

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Problem setup and design requirements

Problem Understanding

The system consists of an inverted pendulum mounted to a motorized cart.

Features of this system:

1. It is unstable without control, i.e., the pendulum will simply fall over if the cart isn't moved to balance it.
2. The dynamics of the system are nonlinear.

Objective:

The objective of the control system is to balance the inverted pendulum by applying a force to the cart that the pendulum is attached to.

Assumption:

We consider a two-dimensional problem where the pendulum is constrained to move in the vertical plane. For this system:

- ▶ The control input is the force F that moves the cart horizontally.
- ▶ The outputs are the angular position of the pendulum θ and

Force analysis and system equations

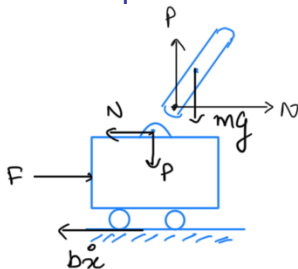


Figure 2: Forces on Inverted Pendulum

Equations of Motion

Pendulum

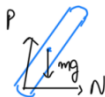


Figure 3: Pendulum

MATLAB representation

Transfer Function

We can represent the transfer functions derived above for the inverted pendulum system within MATLAB employing the following commands.

From input u to output x :

$$P_{cart}(s) = \frac{4.182 \times 10^{-6} s^2 - 1.025 \times 10^{-4}}{2.3 \times 10^{-6} s^4 + 4.182 \times 10^{-7} s^3 - 7.172 \times 10^{-5} s^2 - 1.025 \times 10^{-4} s}$$

From input u to output ϕ :

$$P_{pend}(s) = \frac{1.045 \times 10^{-5} s}{2.3 \times 10^{-6} s^3 + 4.182 \times 10^{-7} s^2 - 7.172 \times 10^{-5} s - 1.025 \times 10^{-4}}$$

Continuous-time transfer function.

State -Space

We can also represent the system using the state-space equations.

The following additional MATLAB commands create the state-space

Data Generation Analysis using Gymnasium

This work uses the InvertedPendulum-v5 environment from the Gymnasium library, which is a MuJoCo-based environment. The task is to balance an inverted pendulum on a cart by applying horizontal forces.

Import

```
import gymnasium as gym
env = gym.make("InvertedPendulum-v5")
```

Action Space

The action space is continuous and represented as:

$$\text{Box}([-3.0], [3.0], (1,), \text{float32})$$

The agent takes a one-dimensional continuous action (force applied to the cart).

Table 1: Action space specification of InvertedPendulum-v5.

Name	Action	Control Mode	Control Mode	Number of Dimensions	...
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Conclusion and Improvements

Conclusion about Random Agent

The four plots paint a clear and consistent picture: a random action strategy is completely inadequate for solving the Inverted Pendulum problem. The agent exhibits no learning or strategic behavior, and its performance relies purely on chance. These results establish a quantitative baseline for failure, against which more advanced control and reinforcement learning approaches can be compared.

Improvements

Here we outline one of the potential improvements that can be employed to achieve meaningful control of the Inverted Pendulum. The current random agent provides a baseline, but it cannot learn or adapt. The following improvements describe how to create a more intelligent agent that learns to balance the Inverted Pendulum effectively.

Implement a Reinforcement Learning Algorithm

The fundamental change is to replace the random action selection